Proof That 10 Is Solitary

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I came to this problem expecting to find a counterexample. Why not, right? The small solitary numbers — 1, 2, 3, 4, 5, 7, 8, 9 — form a sparse set, and the abundancy index 9/5 did not look especially forbidding. Obviously this led us to a proof that closes around a single candidate and locks. Every alternative is killed— one by one— by a constraint that seems far too weak to land the fatal blows. This is the case... until you actually compute through each branch.

0.1 What the Problem Demands

Let $\sigma(n)$ denote the sum of all positive divisors of n. The **abundancy index** of n is $\sigma(n)/n$. A positive integer n is called **solitary** if it is the unique positive integer with its abundancy index. For $n = 10$:

$$\sigma(10) = 1 + 2 + 5 + 10 = 18, \qquad \frac{\sigma(10)}{10} = \frac{9}{5}.$$

The claim: **no integer $m \neq 10$ satisfies $\sigma(m)/m = 9/5$.**

The function σ is multiplicative: if $\gcd(a, b) = 1$, then $\sigma(ab) = \sigma(a)\sigma(b)$. For a prime power, $\sigma(p^e) = (p^{e+1} - 1)/(p - 1)$. If m satisfies $\sigma(m)/m = 9/5$, then $5\sigma(m) = 9m$.

All of this is standard. The difficulty is that 9/5 is a rational number in lowest terms, and abundancy indices carry a rigid denominator-divisibility constraint. But the constraint alone is not enough — it generates an infinite space of candidates when m is factored as $2^a \cdot 5^b \cdot r$ with $\gcd(r, 10) = 1$. We need to show that infinity collapses to finitely many candidates and that all but one die.

0.2 The Denominator Constraint and Why $b = 0$ Fails

The critical tool is the **denominator-divisibility lemma**: if $\sigma(r)/r = A/B$ with $\gcd(A, B) = 1$, then every prime divisor of B divides r. Moreover, if $p \mid B$ and $p \mid r$, writing $r = p \cdot s$ gives

$$\frac{\sigma(s)}{s} = \frac{A \cdot p}{B \cdot \sigma(p)},$$

and the denominator of this new fraction (in lowest terms) must divide s.

Now factor any candidate m as $m = 2^a \cdot 5^b \cdot r$, where $a, b \geq 0$ and $\gcd(r, 10) = 1$. Multiplicativity gives $\sigma(m) = \sigma(2^a)\,\sigma(5^b)\,\sigma(r)$.

Case $b = 0$. Then $\sigma(5^0) = 1$, and $5\sigma(m) = 9m$ becomes

$$5(2^{a+1} - 1)\,\sigma(r) = 9 \cdot 2^a \cdot r,$$

yielding

$$\frac{\sigma(r)}{r} = \frac{9 \cdot 2^a}{5(2^{a+1} - 1)}.$$

The lower bound $\sigma(r)/r \geq 1$ forces $2^a \leq 5$, so $a \in \{0, 1, 2\}$. The three candidates for $\sigma(r)/r$ in lowest terms are 9/5, 6/5, and 36/35. Each denominator is divisible by 5. But r is coprime to 10, so r cannot contain the factor 5 that the denominator-divisibility lemma demands. **No solution with $b = 0$.** This case closes cleanly.

It is the next case where the proof earns its length.

1 The $b \geq 1$ Case: Abundance of Apparent Possibility

When $b \geq 1$, $\sigma(5^b) = (5^{b+1} - 1)/4$. The equation $5\sigma(m) = 9m$ expands to

$$5 \cdot (2^{a+1} - 1) \cdot \frac{5^{b+1} - 1}{4} \cdot \sigma(r) = 9 \cdot 2^a \cdot 5^b \cdot r.$$

Rearrange:

$$\frac{\sigma(r)}{r} = \frac{36 \cdot 2^a \cdot 5^{b-1}}{(2^{a+1} - 1)(5^{b+1} - 1)}. \tag{1}$$

Now impose $\sigma(r)/r \geq 1$. This single inequality does something remarkable: it collapses an infinite (a, b) plane into a finite table. Enumerating all pairs (a, b) with $a \geq 0$, $b \geq 1$ that satisfy (1):

a	b	$\sigma(r)/r$ **(lowest terms)**
0	1	3/2
0	2	45/31
0	3	75/52
0	4	1125/781
0	5	625/434
0	6	28125/19531
0	7	46875/32552
1	1	1/1

For $a = 0$ and $b \geq 8$, the right-hand side of (1) simplifies to

$$\frac{\sigma(r)}{r} = \frac{36 \cdot 5^{b-1}}{5^{b+1} - 1} = \frac{36/5}{5 - 5^{-(b-1)}}.$$

As $b \to \infty$, this approaches $36/25 = 1.44$ from above. Direct computation shows the value drops below 1 for all $b \geq 8$. For $a \geq 1$ and $b \geq 2$, and for $a \geq 2$ and $b = 1$, the inequality $\sigma(r)/r \geq 1$ fails — no further candidates.

I want to pause on what just happened. We started with an equation that, in principle, could have admitted solutions for arbitrarily large a and b — an infinite search. The inequality $\sigma(r)/r \geq 1$ killed all but eight candidate pairs. This is the first crossing point: the object pushes back against the naive expectation that the case analysis would be interminable. It is not. The abundancy index carries a hidden finiteness constraint that the inequality exposes.

The eight that survive are the ones we must now kill.

1.1 Candidates with Immediate Denominator Failure

Four candidates die on first contact with the denominator-divisibility lemma. Since $\gcd(r, 10) = 1$, any candidate whose $\sigma(r)/r$ has denominator divisible by 2 or 5 cannot correspond to any integer r.

- $(0, 1)$: $3/2$, denominator 2 — dead.
- $(0, 3)$: $75/52$, denominator $52 = 2^2 \cdot 13$ — dead.
- $(0, 5)$: $625/434$, denominator $434 = 2 \cdot 7 \cdot 31$ — dead.
- $(0, 7)$: $46875/32552$, denominator $32552 = 2^3 \cdot 13 \cdot 313$ — dead.

The surviving candidates are $(0,2)$, $(0,4)$, $(0,6)$, and $(1,1)$. The last one is the solution $m = 10$: $\sigma(r)/r = 1$ forces $r = 1$, yielding $m = 2^1 \cdot 5^1 = 10$. The other three require deeper elimination.

2 The Hardest Cases

2.0.1 Candidate $(0,2)$

Here $\sigma(r)/r = 45/31$ with prime denominator 31. The lemma forces $31 \mid r$. Write $r = 31 \cdot s$. Then

$$\frac{\sigma(s)}{s} = \frac{45 \cdot 31}{31 \cdot \sigma(31)} = \frac{45}{32}.$$

The denominator $32 = 2^5$ must divide s, contradicting $\gcd(s, 10) = 1$. Dead.

The reasoning is swift, but notice why it works: the denominator-divisibility lemma is iterable. It does not merely reject a candidate — it chases the contradiction into smaller integers. This candidate fails not because 45/31 is impossible as an abundancy index, but because the denominator's prime factor 31 triggers a cascade. The number 31 is unremarkable; what matters is that $\sigma(31) = 32$ introduces a factor of 2 into the denominator of the reduced fraction at the next level. The cascade is sensitive to the *specific* prime, not just its size.

2.0.2 Candidate $(0,4)$

$\sigma(r)/r = 1125/781$. The denominator $781 = 11 \cdot 71$. We must consider both prime factors.

Branch 1: $11 \mid r$. Write $r = 11 \cdot s$.

$$\frac{\sigma(s)}{s} = \frac{1125 \cdot 11}{781 \cdot 12} = \frac{375}{284} = \frac{375}{2^2 \cdot 71}.$$

Denominator now contains 2^2. But $\gcd(s, 10) = 1$ forbids $2 \mid s$ — dead.

Branch 2: $71 \mid r$. Write $r = 71 \cdot s$.

$$\frac{\sigma(s)}{s} = \frac{1125 \cdot 71}{781 \cdot 72} = \frac{125}{88} = \frac{125}{2^3 \cdot 11}.$$

Denominator $88 = 2^3 \cdot 11$ again forces a factor of 2 into s — dead.

Both branches die. The candidate is eliminated.

There is a structural observation worth making explicit: the branching here is not parallel. The lemma does not tell us *which* prime divisor of the denominator initiated the cascade — we must try both. In every branch, the cascade terminates in a contradiction within one step. It is tempting to conjecture that this always happens for the solitary numbers, but I do not know whether it holds in general. The proof for 10 works because the cascades are uniformly short.

2.0.3 Candidate $(0,6)$

$\sigma(r)/r = 28125/19531$. The denominator 19531 is prime. Write $r = 19531 \cdot s$.

$$\frac{\sigma(s)}{s} = \frac{28125 \cdot 19531}{19531 \cdot 19532} = \frac{28125}{19532} = \frac{28125}{2^2 \cdot 19 \cdot 257}.$$

The denominator contains 2^2 — contradiction. Dead.

2.1 The Proof Closes

Every candidate has been examined:

The only solution to $\sigma(m)/m = 9/5$ is $m = 10$. Therefore 10 is solitary.

The proof's engine is the iterated denominator-divisibility lemma, applied across a finite candidate table generated by the inequality $\sigma(r)/r \geq 1$. The inequality prunes infinity to finiteness; the lemma prunes finiteness to uniqueness. Both are necessary. Neither alone suffices.

Candidate	**Fate**
$(0,1)$	Immediate denominator failure (2)
$(0,2)$	Cascade contradiction via $\sigma(31) = 32$
$(0,3)$	Immediate denominator failure (2)
$(0,4)$	Both branches cascade to factor 2
$(0,5)$	Immediate denominator failure (2)
$(0,6)$	Cascade contradiction via $\sigma(19531)$
$(0,7)$	Immediate denominator failure (2)
$(1,1)$	Survives: $m = 10$

3 What This Proof Leaves Open

The proof I expected to find would have been a direct application of a general theorem: "if n satisfies condition C, then n is solitary." No such theorem exists. The solitude of each small number is proved by a bespoke argument — different denominators, different cascades, different branch structures. What unifies them is the denominator-divisibility lemma, but that lemma is a tool, not a theory.

The proof for 10 succeeds because the candidate table is small and every cascade terminates in a single step. For larger solitary numbers, the candidate tables grow and cascades can run longer. The solitude of 10 is not deep in the way that, say, the solitude of a prime power is deep. It is deep in a different way: the proof works because $10 = 2 \cdot 5$ is exactly small enough that the finite set of candidates is manageable by hand. The structural question is whether there exists a single solitary number whose proof requires a cascade of length greater than 1 — a cascade where one iteration of the lemma does not suffice, and the contradiction is genuinely deferred.

If such a number exists, its proof would be qualitatively different from the proof for 10. The argument would need to track not just whether a denominator contains a forbidden prime but whether the cascade eventually forces one — a property of the prime's orbit under σ, which is not well understood. The smallest candidate for such behavior might be a number whose abundancy index generates a denominator with a prime p such that $\sigma(p)$ itself has a prime factor forbidden by coprimality — and finding such a p is not a matter of computation but of structural necessity.

I do not know whether that number exists. The proof above does not need it to exist — it only needs the eight candidates for 10 to die.

Structural type: $\langle D_\infty;\ T_{\bowtie};\ R_{\leftrightarrow};\ P_{\pm};\ F_{\hbar};\ K_{slow};\ G_{\aleph};\ \Gamma_{seq};\ \Phi_c;\ H_2;\ n{:}m;\ \Omega_{\mathbb{Z}_2}\rangle$